

Prolipoprotein Diacylglyceryl Transfer by Lgt: A Review-Style Synthesis

Scope module: *Pseudomonas putida* KT2440 UniPathway UPA00664 — Lgt (PP_5142 / UniProtKB Q88CN8; EC 2.5.1.145; Rhea:RHEA:56712). Search date: 2026-07-11 (PubMed).

1. Executive summary

Bacterial lipoproteins are membrane-anchored proteins whose defining feature is a lipid-modified N-terminal cysteine. The **first and committed step** in building that anchor is catalyzed by **prolipoprotein diacylglyceryl transferase (Lgt)**, an integral inner-membrane enzyme that transfers a **diacylglyceryl group from the membrane phospholipid phosphatidylglycerol (PG)** onto the thiol of the invariant cysteine of a "lipobox," producing a thioether-linked **S-diacylglyceryl-prolipoprotein** (EC 2.5.1.145; RHEA:56712)

([P 28698362](#));

[P 26729647](#)).

This review treats the **Lgt reaction itself** as the biological system of interest — a single, mechanistically discrete, membrane-embedded acyl/glyceryl-transfer event — and deliberately separates it from the downstream signal-peptide cleavage (LspA) and N-acylation (Lnt) steps, and from the still-later Lol sorting machinery, which are frequently conflated with it.

The key conclusions are: (i) Lgt performs the committed, gatekeeping reaction of the entire lipoprotein-maturation pathway and imposes a hard ordering constraint on everything downstream

([P 24780125](#));

[P 42126242](#));

(ii) it is **universally present in bacteria and absent from archaea**, marking it as a bacterial-specific innovation tied to bacterial phospholipid chemistry

([P 42126242](#));

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(iii) its **essentiality is lineage-dependent** — lethal in Proteobacteria, dispensable in several monoderms

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and (iv) crystal structures rationalize a **membrane-lateral, two-site catalytic mechanism** with essential active-site arginines

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Because it is essential in Gram-negatives, surface-accessible, and mechanistically distinct from human enzymes, Lgt is an actively pursued antibiotic target

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2. Definition and biological boundaries

2.1 What is included

The system is the **single enzymatic reaction**:

phosphatidylglycerol + [prolipoprotein]-L-cysteine → [prolipoprotein]-S-1,2-diacyl-sn-glycerol-L-cysteine + sn-glycerol-1-phosphate

catalyzed by Lgt. Its scope comprises: (1) recognition of an unmodified prolipoprotein bearing a **lipobox** — the consensus [LVI]-[ASTVI]-[GAS]-C motif at the signal-peptide/mature-protein junction — as it emerges laterally from the translocon; (2) binding and activation of the PG donor; and (3) transfer of the diacylglycerol moiety to the cysteine sulfhydryl to form a

thioether

bond

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The *P. putida* representative Q88CN8/PP_5142 is a canonical member of this reviewed family.

2.2 What should be treated separately (common confusions)

- **LspA (signal peptidase II)** cleaves the signal peptide *after* Lgt acts, unmasking the α -amino group of the now N-terminal S-diacylglyceryl-cysteine. This is a peptidase reaction, mechanistically and structurally unrelated to Lgt

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- **N-acylation (Lnt or non-orthologous equivalents)** adds the **third** acyl chain to the α -amino group, generating **triacylated** lipoproteins. Gram-negatives use **Lnt**; several Gram-positives that lack an *E. coli*-type Lnt achieve tri-acylation by other enzymes (e.g., Actinobacterial **Ppm2**), so di- versus tri-acylation and the identity of the acyltransferase are all **downstream of, and separable from, Lgt**

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- **The Lol pathway (LolCDE/LolA/LolB)** and the **" +2 rule "** govern *trafficking/sorting* of already-mature lipoproteins to the outer membrane; these are localization events, not modification chemistry, and act well downstream of Lgt

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- **TLR2 immune recognition** of di-/tri-acylated lipopeptides is a *host* consequence of the modification, not part of the enzymatic system

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- **Archaeal "lipoprotein" lipidation** is a convergent but molecularly distinct process with no Lgt homolog

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2.3 Competing definitions

There is little dispute over the reaction chemistry. Two definitional tensions exist: (a) whether Lgt is a true **polytopic seven-transmembrane** protein or a **seven-helix bundle peripherally embedded in the inner leaflet** — a point raised by solubilization/behavioral evidence

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 that qualifies the "integral membrane" label of the crystallographic model
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and (b) the loose colloquial use of "lipoprotein maturation" to mean the whole Lgt→LspA→Lnt cascade, versus the strict UniPathway UPA00664 module, which records only the Lgt transfer step.

3. Mechanistic overview

Best current model (ordered, membrane-embedded catalysis):

1. **Substrate presentation.** A prelipoprotein is exported through the cytoplasmic membrane, predominantly via **Sec** (occasionally **Tat**). As it exits the translocon, its hydrophobic signal peptide with the C-terminal lipobox remains membrane-associated and is recognized by Lgt

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2. **Donor loading.** Lgt binds **phosphatidylglycerol** in a dedicated site. Structures of E. coli Lgt captured PG and, separately, the inhibitor **palmitic acid**, defining **two binding sites** (a PG/donor site and a lipid/acyl site)

([P 26729647](#)).

A companion mechanistic study describes how the **PG headgroup is activated** for transfer

([P 31340643](#)).

3. **Diacylglyceryl transfer.** The **thiol of the invariant lipobox cysteine** attacks the sn-1/sn-2 diacylglyceryl unit of PG, forming a **thioether-linked S-diacylglyceryl-cysteine** and releasing sn-glycerol-1-phosphate. **Arg143 and Arg239 are essential** for this transfer (likely stabilizing the phosphate/transition state and orienting substrates)

([P 26729647](#)).

4. **Product release.** The lipid-modified lipobox peptide **exits laterally** into the bilayer, consistent with a laterally open active site rather than an aqueous channel

([P 26729647](#)).

The S-diacylglyceryl-prolipoprotein is then handed to LspA.

Obligatory vs conditional vs accessory steps. - *Obligatory*: PG binding, cysteine thioether formation. Without Lgt action, LspA cannot cleave — establishing a **strict upstream dependency**

([P 24780125](#)).

- *Conditional/lineage-specific*: subsequent **N-acylation by Lnt** (present in diderms and some monoderms, historically thought absent in many Firmicutes but now known to occur via alternative enzymes in some Gram-positives). - *Accessory*: Sec vs Tat choice for export; downstream Lol sorting and the +2 / +3,+4 retention rules

([P 17350956](#)).

4. Major molecular players and active assemblies

Player	Role in the Lgt step	Evidence
Lgt (PP_5142/Q88CN8 in <i>P. putida</i>)	Catalyzes diacylglyceryl transfer; ~7-helix inner-membrane enzyme; conserved catalytic core plus variable "arm" and "head" domains	P 26729647 ; P 42126242
Phosphatidylglycerol (PG)	Diacylglyceryl donor ; anionic membrane phospholipid	P 26729647 ; P 31340643
Prolipoprotein + lipobox Cys	Acceptor substrate; [LVI][ASTVI][GAS]C motif	P 28698362 ; P 28506828
Arg143, Arg239 (<i>E. coli</i> numbering)	Essential catalytic/positioning residues	P 26729647
Sec (and Tat) translocon	Delivers prelipoprotein to the membrane for Lgt	P 32490790 ; P 24780125

Lgt is generally described as a **monomeric** enzyme; chromatographic behavior as an active monomer has been reported, alongside the debate on the strength of its membrane association ([P 31256204](#)).

No stable multi-subunit "maturation complex" is required for the Lgt step itself; the three enzymes act **sequentially** rather than as a fixed holoenzyme ([P 28698362](#)).

5. Evolutionary and cell-biological variation

5.1 Across lineages

- **Universality and origin.** Lgt is found in **all bacteria** and **absent from archaea**

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The catalytic core is deeply conserved; the peripheral **"arm" and "head" domains vary** and appear to underlie functional diversity among pathogens

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— these variable modules are the plausible "later elaborations," while the PG-binding catalytic core represents the **ancestral, best-representative** unit.

- **Diderms vs monoderms and alternative N-acylation routes.** Both use Lgt then LspA. **Lnt-mediated tri-acylation** is characteristic of diderms, but the downstream N-acylation machinery is strikingly **non-orthologous** across lineages while the Lgt step stays invariant: low-GC Firmicutes such as *S. aureus* **lack an E. coli-type Lnt yet still make triacylated lipoproteins** (e.g., α -N-acylated SitC), implying a distinct acyltransferase

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and Actinobacteria (*Corynebacterium/Myobacterium*) N-acylate via **Ppm2** in the *ppm* operon

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This is a clear case of the same outcome reached by different molecular means — and underscores that only the **Lgt diacylglyceryl transfer** is universal and mechanistically fixed

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- **Sorting downstream diverges.** *E. coli* uses the **"+2 rule"** (Asp at +2 → inner-membrane retention), whereas *P. aeruginosa* follows a **"+3,+4 rule"**

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and **Borrelia** lipoproteins default to surface display

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P 16468989
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These are downstream of Lgt but illustrate lineage plasticity of the broader system.

5.2 Physiological states and consequences

- In **Proteobacteria**, Lgt is required for outer-membrane integrity; its loss permeabilizes the OM and sensitizes to serum/antibiotics

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- In **Gram-positive pathogens**, Lgt output (lipoproteins) shapes **host innate immunity via TLR2**, with double-edged effects on virulence and inflammation

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- **Archaea** solve the analogous problem with a **distinct, non-Lgt machinery** dictated by isoprenoid ether lipids

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6. Constraints, dependencies, and failure modes

- **Ordering constraint (hard)**. Lgt must act **before** LspA; LspA needs the S-diacylglycerol substrate. Thus Lgt is a rate/route-defining gate

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- **Compartment constraint**. Reaction occurs at the **cytoplasmic (inner) membrane**, using membrane PG and a membrane-tethered substrate; substrate/product enter and leave **laterally**

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- **Substrate specificity.** An **invariant cysteine** in a recognizable lipobox is required; the sulfhydryl is the sole nucleophile — no cysteine, no thioether, no downstream maturation

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P 28698362;
P 28506828).

- **Failure modes.** Loss/inhibition of Lgt → accumulation of unmodified prolipoproteins, OM permeabilization, morphological defects and death in Proteobacteria (independent of Lpp)

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P 35938851);
 attenuated virulence and altered TLR2 signaling in Gram-positives
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P 31226163).

- **What is ruled out.** LspA-first or Lnt-first routes are excluded by the substrate requirements of those enzymes; a purely cytoplasmic/aqueous reaction is excluded by the membrane-lateral structural model and PG dependence

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P 26729647;
P 24780125).

7. Controversies and open questions

1. **Membrane topology.** Is Lgt a genuine 7-TM integral protein (crystallographic view,

P 26729647)
 or a 7-helix bundle **peripherally embedded** in the inner leaflet (solubilization/behavioral and MODA evidence,

P 31256204)?
 This bears directly on inhibitor design.

2. **Precise chemistry of PG activation** and the exact catalytic roles of Arg143/Arg239 (electrostatic transition-state stabilization vs substrate positioning) remain to be fully resolved at transition-state resolution

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P 31340643
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3. **Essentiality determinants.** Why is Lgt essential in Proteobacteria but dispensable in several monoderms

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Is essentiality strictly tied to OM lipoprotein cargo (e.g., Lol substrates), or to specific critical lipoproteins?

4. **Cross-organism extrapolation.** Much mechanistic detail comes from *E. coli*; functional divergence via the variable arm/head domains cautions against assuming identical behavior in, e.g., *P. putida*, mycobacteria, or Firmicutes

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5. **Druggability & selectivity.** Tool inhibitors exist (e.g., G2824, palmitic acid as a co-crystallized ligand), and Lgt inhibition is insensitive to Lpp-deletion resistance

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but selective, whole-cell-active leads and their resistance liabilities are still open

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P 31036406
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8. Key references

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P 42126242

— Lgt universal in bacteria, absent in archaea; conserved core + variable domains.

- Mao, Zhao, Kang, et al. *Crystal structure of E. coli lipoprotein diacylglycerol transferase*. Nat Commun 2016.

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- structures with PG and palmitic acid; Arg143/Arg239; lateral access.
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- PG activation mechanism.
- Noland, Kattke, Diao, et al. *Structural insights into lipoprotein N-acylation by Lnt*. PNAS 2017.

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- pathway ordering; lipobox recognition; Lgt→LspA→Lnt.
- Zückert. *Secretion of bacterial lipoproteins...* 2014.

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- pathway ordering, monoderm vs diderm, Lol/+2 rule.
- Legood, Seng, Boneca, Buddelmeijer. *A Defect in Lipoprotein Modification by Lgt Leads to... Cell Death in E. coli... Independent of Lpp*. 2022.

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- essentiality/failure mode in Proteobacteria.
- Dautin, Argentini, Mohiman, et al. *Role of the unique, non-essential Lgt in Corynebacterium*. 2020.

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- Diao, Komura, Sano, et al. *Inhibition of E. coli Lgt Is Insensitive to Resistance Caused by Deletion of Braun's Lipoprotein*. 2021.

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- first Lgt inhibitor (G2824); OM permeabilization.
- Sangith, Kumar, Sankaran. *Evidence to Suggest Lgt is a Weakly Associated Inner Membrane Protein*. 2019.

P 31256204

- topology controversy.

- El Arnaout & Soulimane. *Targeting Lipoprotein Biogenesis: Considerations towards Antimicrobials*. 2019.

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— druggability of Lgt/LspA/Lnt.

- Hong, Makarova, Xu, et al. *...first biogenesis machinery of archaeal lipoproteins*. 2024.

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— archaeal absence of Lgt; distinct route.

- Narita & Tokuda. *Amino acids at positions 3 and 4 determine membrane specificity of P. aeruginosa lipoproteins*. 2007.

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Schmaler et al. 2010.

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— Gram-positive lgt dispensability, TLR2 biology.

- Asanuma, Kurokawa, et al. *Structural evidence of α -aminoacylated lipoproteins of Staphylococcus aureus*. 2011.

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— Lnt-independent tri-acylation in a Firmicute.

- Mohiman, Argentini, et al. *The ppm operon is essential for acylation and glycosylation of lipoproteins in Corynebacterium glutamicum*. 2012.

P 23029442

— Ppm2 as an alternative N-acyltransferase in Actinobacteria.

- Sangith & Sankaran. *Stringency of LspA in recognition of signal peptides*. 2017.

P 28506828

— lipobox definition, LspA boundary.

Uncertainty note: Most atomic-resolution mechanism derives from *E. coli* Lgt; extrapolation to *P. putida* (Q88CN8) and to Gram-positive/actinobacterial homologs is inferred from strong sequence/structure conservation of the catalytic core but is qualified by documented divergence in peripheral domains and in downstream steps (di- vs tri-acylation; sorting rules). Direct experimental characterization of *P. putida* KT2440 Lgt (PP_5142) is sparse in the primary literature; as a diderm Gammaproteobacterium closely related to *P. aeruginosa* and *E. coli*, it is expected to require Lgt for outer-membrane lipoprotein biogenesis and to follow the "+3,+4" *Pseudomonas* sorting logic downstream (P 17350956), but this remains an inference rather than a direct measurement.

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